



Scottish Graduate School of Social Science

Sgoil Cheumnaichean Saidheans
Sòisealta na h-Alba

Collecting Digital Data

What Are APIs and Why Use Them?

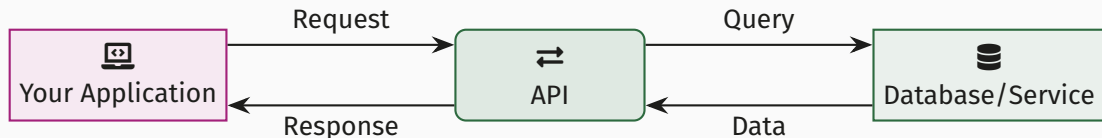
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24 February 2026

What Are APIs and Why Use Them?

What is an API?



API = Application Programming Interface

Think of it as a **translator**: your application doesn't need to know *how* the database is organised — it just asks the API in a standard way, and the API returns structured data.

APIs vs Web Scraping

	Web Scraping	APIs
Data format	Unstructured HTML	Structured JSON/XML
Reliability	Breaks when site changes	Versioned, documented
Legality	Grey area	Usually explicit ToS
Rate limits	You must self-impose	Enforced by provider
Authentication	Usually none	Often API key

Rule of thumb

If an API exists, use it. Web scraping is a last resort.

Anatomy of an API Call

`https://data.police.uk/api/stops-force?force=city-of-london&date=2024-01`

Base URL

Endpoint

Query parameters

key=value pairs joined by &

✉ Headers

Accept: application/json

🔑 Authentication

API key, OAuth token, or none

The UK Police API is **open** — no authentication needed. Many other APIs require an API key sent as a header or query parameter.

What to look for when you open the docs:

- ✓ **Endpoints** — what URLs are available and what data do they return?
- ✓ **Authentication method** — API key, OAuth, or open access?
- ✓ **Rate limits** — how many requests per minute/day?
- ✓ **Response format** — JSON, XML, CSV?
- ✓ **Example requests & responses** — try them in a browser first!

UK Police API — `data.police.uk/docs/`

- ✓ Open access (no key needed)
- ✓ Returns JSON
- ✓ Rate limit: 15 requests per second
- ✓ Endpoints: stop-and-search, crimes, neighbourhoods, forces

JSON Deep-Dive

```
{
  "type":  "Person",
  "age_range":  "18-24",
  "legislation":  [
    {
      "name":  "John Doe",
      "age":  21
    }
  ],
  "outcome":  true
}
```

— Keys — Values

JSON types:

- **Object** { } — key-value pairs
- **Array** [] — ordered list
- **String, Number, Boolean, null**

JSON is the most common format for API responses. Each record is a set of **key-value** pairs. Values can be strings, numbers, booleans, arrays, or nested objects.

Research in Action: APIs in Social Science

King, Pan & Roberts (2013)

How Censorship in China Allows Government Criticism but Silences Collective Expression. American Political Science Review.

Used social media APIs to collect millions of posts from Chinese platforms before and after censorship, revealing that the government targets collective action, not criticism per se.

Barberá (2015)

Birds of the Same Feather Tweet Together: Bayesian Ideal Point Estimation Using Twitter Data. Political Analysis.

Used the Twitter API to collect follower networks and estimate the political ideology of millions of users, demonstrating that API data can scale political science measurement.

Takeaway

APIs provide structured, reproducible access to large-scale digital data that would be impossible to collect manually.

Practical 2: UK Police API

- Make your first API request (in a browser, then in code)
- Parse JSON responses into tabular data
- Explore stop-and-search data for a chosen police force
- Handle pagination and query parameters

Open the **Practical 2** notebook in Google Colab
(Python or R — your choice!)